Skin Cancer Detection: From CNNs to Swin Transformers

A Rigorous Architectural Analysis

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Abstract

Skin cancer is a highly prevalent malignancy where early diagnosis is critical. Manual dermoscopy, however, suffers from severe inter-observer variability. We present an automated classification pipeline using the HAM10000 dataset, evaluating baseline CNNs through to the Swin Transformer. By leveraging Shifted Window Self-Attention, integrating a "Not a Lesion" healthy skin class, minority oversampling, and AdamW with Decoupled Weight Decay, our Swin-T model achieves 92% validation accuracy, providing a robust, real-time solution for automated dermatological diagnosis.

1 Motivation

Skin cancer incidence rates are rising globally. While early diagnosis drastically improves prognosis, standard manual dermoscopy is time-consuming and requires specialized expertise. This reliance on human interpretation introduces severe inter-observer variability and high false-negative rates for rare malignancies. There is an urgent clinical need for automated, mathematically rigorous, and highly sensitive real-time diagnostic tools to assist dermatologists and improve patient outcomes.

2 Problem Definition

The core challenge is the automated classification of distinct skin lesion types from dermoscopic images using the HAM10000 dataset. Key Constraints:

- Extreme Imbalance: The dominant class (nv) constitutes ≈67% of samples, while rare malignant classes represent $< 2\%$.
- High Sensitivity: The network must identify critical malignancies (melanoma, basal cell carcinoma) with minimal false negatives.
- Latency: Inference must occur in under 2 seconds.

3 Key Contributions

We engineered a robust pipeline bridging local feature extraction and global context modeling:

- Architectural Evolution: Evaluated baseline CNNs, ResNet-50, EfficientNet, YOLOv4, and standard ViT. We ultimately deployed the Swin Transformer for its $\mathcal{O}(M^2hwC)$ linear efficiency via Shifted Window Self-Attention.
- Class Balancing: Mitigated dataset skew by introducing a custom 8th class ("Not a Lesion") using ISIC 2020 healthy skin data and systematic minority oversampling.
- Optimization Strategy: Utilized a rigorous two-phase fine-tuning protocol using the AdamW optimizer to apply true L_2 penalties isothermally.

4 High-Level Results & Analysis

Our methodological interventions resulted in significant performance gains. Standard CNNs suffered from severe overfitting due to spatial limitations, while global ViT architectures were computationally restricted. The Swin Transformer effectively synthesized hierarchical local extraction with global attention.

Table 1: Model Performance Comparison

Model	Train Acc.	Val Acc.	Params
CNN (scratch)	70%	62%	~2M
ResNet-50	85%	76%	25.6M
EfficientNet-B3	90%	82%	11.1M
YOLOv4	88%	79%	–
ViT-B/16	91%	83%	86M
Swin-T (Ours)	97%	92%	29M

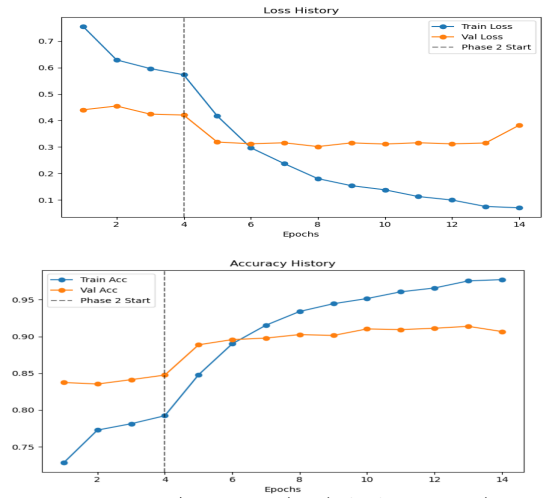

Figure 1: Training/Val Loss (top) & Accuracy (bottom).

The Swin-T model achieved a state-of-the-art 92% validation accuracy. The inclusion of the "Not a Lesion" class successfully prevented false positives on healthy skin, while our optimization dynamics ensured robust sensitivity for critical malignancies.